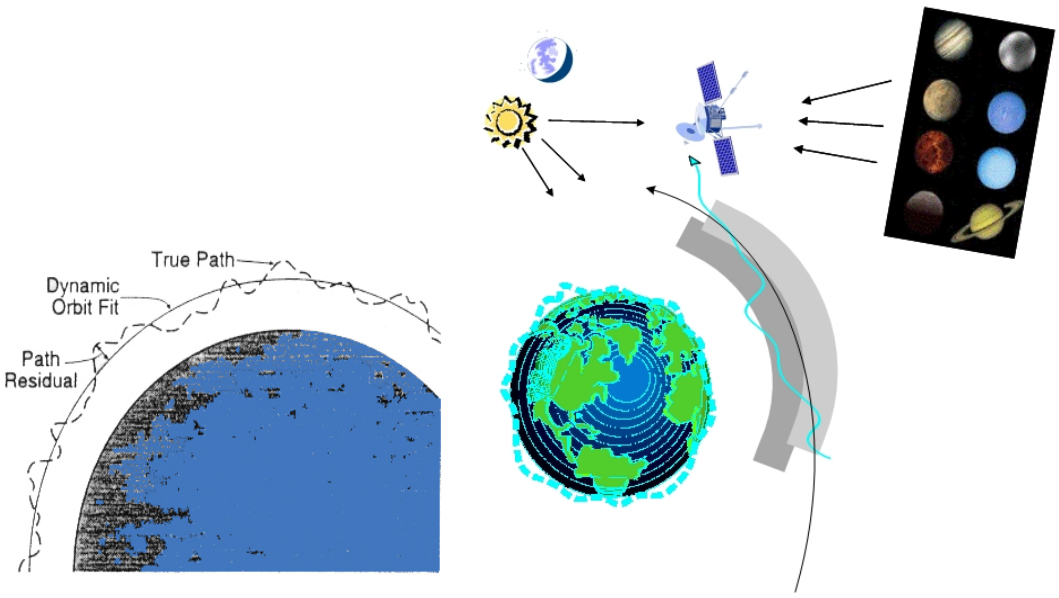
Chapter 8. Special Perturbation Techniques

8.1 Historical Background

- Perturbation Analysis is ultimately for the purpose of Orbit Propagation
  - Numerical (Special Perturbation), Analytical (General Perturbation), Semi-Analytical
  - The most accurate way is numerical
    - Encke's Formulation
    - Cowell's Formulation

- Perturbations are deviations from a normal, idealized, or unperturbed motion
  - Tend to view the universe as highly regular and predictable
  - Yet accurate observational data often reveals unexplained irregularities of motion superimposed on more regular average/mean motions of the celestial bodies

  - Idealized solutions only approximate the observed motion
  - Actual motion will vary from the theoretical two-body path due to perturbations caused by
    - Other Bodies: Sun, Moon, etc.
    - Additional Forces not considered in Keplerian motion: Non-Spherical Central Body, Air Drag, Solar Radiation, (Thrust), etc.



8.2 Introduction to (Special) Perturbations

- Special Perturbation Techniques
  - **Numerically** integrate the equations of motion including all necessary perturbing accelerations
  - Produces a specific/special answer that is valid only for the given data:
    - Initial Conditions
    - Force-Model Parameters
  - Advantages : Accurate Results, Easy Implication
  - Disadvantages : New data leads to New Integration Errors (Round-off, Truncation)

- Conservative vs. Non-conservative Perturbation Forces
  - Conservative
    - Central-Body/Non-Spherical/Third-Body Gravitational Effects
    - Total Energy for Conservative-Force System is Constant
  - Non-Conservative
    - Solar-Radiation Pressure, Air Drag, Thrust
    - May lose/gain Energy mainly through friction or external sources

- Potential Function (related to Potential Energy)
  - One way to mathematically characterize Conservative Forces
  - Its Gradient (or Directional Derivatives) yields Acceleration Component

  - Disturbing Function is simply the difference between Perturbed and Unperturbed Potential Function
    - Disturbing Force simply expresses the specific force (acceleration) that is Perturbing Two-Body motion
    - For example of the Spherical and Aspherical Potentials

$$U = U_{2\text{-}body} + R$$

U : Potential Function

$U_{2\text{-}body}$: Spherical Central Body's Potential Function

R : Disturbing Potential Function
(represents only the perturbing effect of Non-Spherical terms)

- Representation of (Perturbed) Two-Body Motion
  - Spherical Central Body's (=Nominal=Unperturbed) Potential Function

$$U_{2-body} = \frac{\mu}{r}$$

$$\Rightarrow \nabla U_{2-body} = \frac{\partial U_{2-body}}{\partial x}\hat{i} + \frac{\partial U_{2-body}}{\partial y}\hat{j} + \frac{\partial U_{2-body}}{\partial z}\hat{k}$$

$$where \;\; \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \;,\; r = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow \nabla U_{2-body} = -\frac{1}{2}\frac{\mu}{\left(\sqrt{x^2 + y^2 + z^2}\right)^3}\left[2x\hat{i} + 2y\hat{j} + 2z\hat{k}\right]$$

$$= -\frac{\mu}{r^3}\vec{r}$$

- Can express Two-Body equations of motion compactly as

$$\ddot{\vec{r}}_{2-body} = \nabla U_{2-body} \tag{8-1}$$

- Addition of Conservative Perturbation Forces to Two-Body Motion

$$\ddot{\vec{r}} = \nabla U, \quad U = U_{2-body} + R$$

The specific form of R depends on the particular type of perturbing forces: Aspherical Central Body, Third-Body Gravitational Forces

8.3 Encke's Formulation

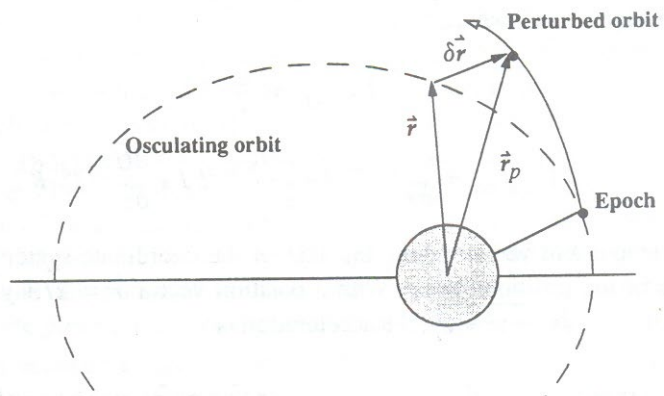


Figure 8-1. **Encke's Method.** The foundation for Encke's method relies on the difference between the osculating (two-body) orbit and the actual perturbed motion, shown as δr. By integrating this difference, the method can save many computations.

- Encke's Method
  - Numerically integrate
    differences of the actual perturbed motion from the osculating (reference) orbit rather than the complete attraction and specific forces involved
  - Reference motion from departure is calculated by unperturbed two-body motion

- Obtain enough accuracy with limited computing abilities
- Not very popular today because modern computing facility permits a complete numerical solution

$$\vec{r}_p = \vec{r} + \delta\vec{r}$$
$$\dot{\vec{r}}_p = \dot{\vec{r}} + \delta\dot{\vec{r}}$$
$$\ddot{\vec{r}}_p = \ddot{\vec{r}} + \delta\ddot{\vec{r}}$$

Two-Body Acceleration: $\ddot{\vec{r}} = -\dfrac{\mu}{r^3}\vec{r}$

Perturbed Acceleration: $\ddot{\vec{r}}_p = -\dfrac{\mu}{r_p{}^3}\vec{r}_p + \vec{a}_p$

Difference between Nominal (Unperturbed) and Perturbed Accelerations:

$$\begin{aligned}
\delta\ddot{\vec{r}} &= \ddot{\vec{r}}_p - \ddot{\vec{r}} \\
&= \vec{a}_p + \mu\left(\frac{\vec{r}}{r^3} - \frac{\vec{r}_p}{r_p{}^3}\right) \\
&= \vec{a}_p + \frac{\mu}{r^3}\left(\vec{r} - \frac{r^3}{r_p{}^3}\vec{r}_p\right) \\
&\qquad \left(\Uparrow\ \vec{r} = \vec{r}_p - \delta\vec{r}\right) \\
&= \vec{a}_p + \frac{\mu}{r^3}\left[\left(1 - \frac{r^3}{r_p{}^3}\right)\vec{r}_p - \delta\vec{r}\right]
\end{aligned}$$

$1 - \dfrac{r^3}{r_p{}^3}$: Appearance of the difference between two nearly equal quantities

$$\begin{aligned}
\vec{r} &= \vec{r}_p - \delta\vec{r} \\
r^2 &= \vec{r}\cdot\vec{r} \\
&= (\vec{r}_p - \delta\vec{r})\cdot(\vec{r}_p - \delta\vec{r}) \\
&= r_p^2 - 2\vec{r}_p\cdot\delta\vec{r} + \delta\vec{r}\cdot\delta\vec{r} \\
&\left(\Uparrow\ q \equiv \frac{-2\delta\vec{r}\cdot\vec{r}_p + \delta\vec{r}\cdot\delta\vec{r}}{r_p^2}\right) \\
&= (1+q)r_p^2 \\
\Rightarrow \frac{r^2}{r_p^2} &= 1 + q \\
\Rightarrow \frac{r^3}{r_p^3} &= (1+q)^{3/2}
\end{aligned}$$

$$
\begin{aligned}
1-\frac{r^3}{r_p^3} &= 1-(1+q)^{3/2} \\
&= \left[1-(1+q)^{3/2}\right]\frac{\left[1+(1+q)^{3/2}\right]}{\left[1+(1+q)^{3/2}\right]} \\
&= \frac{1-(1+q)^3}{\left[1+(1+q)^{3/2}\right]} \\
&\left(\Uparrow\ (1+q)^3 = 1+q^3+3q^2+3q\right) \\
&= \frac{-\left[3q+3q^2+q^3\right]}{\left[1+(1+q)^{3/2}\right]} \\
&\equiv f(q)
\end{aligned}
$$

Therefore $\delta\ddot{\vec{r}}$ equation can be expressed as follows:

$$
\delta\ddot{\vec{r}} = \vec{a}_p + \frac{\mu}{r^3}\left[f(q)\vec{r}_p - \delta\vec{r}\right],\ \delta\vec{r}(t_0) = \delta\dot{\vec{r}}(t_0) = 0
$$

$$
where\ \vec{r}_p = \vec{r} + \delta\vec{r}
$$

$$
q = \frac{\delta\vec{r}\cdot\delta\vec{r} - 2\delta\vec{r}\cdot\vec{r}_p}{r_p^2}
$$

$$
f(q) = \frac{-q[3+3q+q^2]}{[1+(1+q)^{3/2}]}
$$

$\vec{r}$ is computed from the Two-Body solution

$\delta\vec{r}$ and $\delta\dot{\vec{r}}$ are obtained by numerically integrating the above $\delta\ddot{\vec{r}}$ equation

When $\delta\vec{r}$ and $\delta\dot{\vec{r}}$ become too large, rectify the osculating orbit

8.4 Cowell's Formulation

- Common to use Cowell's formulation to set up the equations of motion for numerical integration
  - (Numerically) Integrate all perturbations and primary acceleration together
  - One of the most attractive characteristics for numerical techniques is the ability to incorporate any arbitrary disturbing acceleration

$$
\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} + \vec{a}_{perturbed} \tag{8-3}
$$

where $\vec{a}_{perturbed}$ is the total acceleration caused by other forces acting on the satellite, i.e., additional effects on the original two-body equation

- Common to recast three (scalar) second-order differential equations of motion into six (scalar) first-order differential equations
- This first-order system is known as Variation of Cowell's formulation.

$$\vec{X} = \begin{bmatrix} x \\ y \\ z \\ v_x \\ v_y \\ v_z \end{bmatrix} \quad , \quad \dot{\vec{X}} = \begin{bmatrix} v_x \\ v_y \\ v_z \\ -\frac{\mu}{r^3}x + a_{px} \\ -\frac{\mu}{r^3}y + a_{py} \\ -\frac{\mu}{r^3}z + a_{pz} \end{bmatrix}$$

8.5 Numerical Integration Methods

Runge-Kutta method

(single step, multi-step, fixed step, variable step size, etc)

8.6 Disturbing Forces

- Numerical integration of Cowell's formulation requires to obtain a mathematical model for each perturbing force

8.6.1 Gravity Field of a Central Body

- The gradient of the Spherical Potential function (for a spherical central body) yields the (nominal) two-body acceleration
  - Need to derive the Aspherical Potential Function that includes the perturbing accelerations due to a non-spherical central body

8.6.1.1 Derivation of Aspherical Potential Function

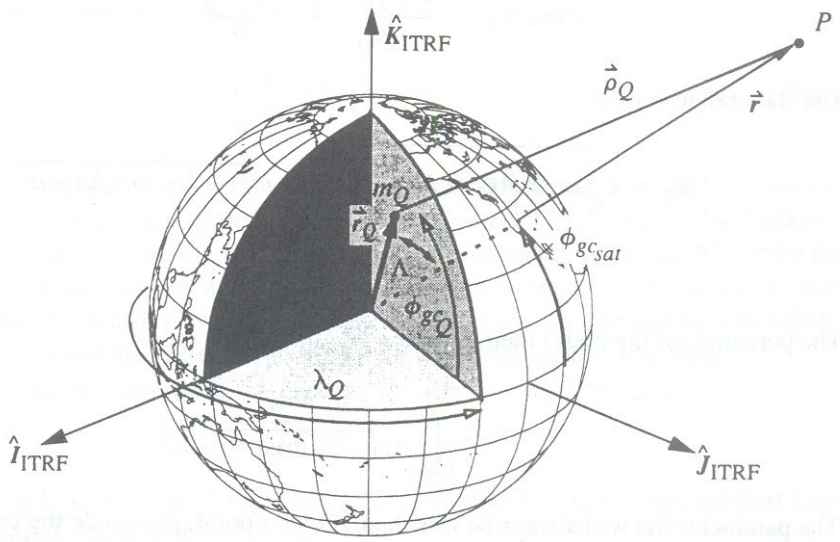


Figure 8-3. Deriving the Gravitational Potential . We consider each point in the Earth, m_Q, when determining the gravitational potential at P. The latitude is a geocentric value. Λ is the angle between the vectors r_Q and r .

- Derivation of Aspherical Potential Function by Integration

  An infinite number of masses m_Q at points Q

  The potential per unit mass at point P, which is denoted by U, is the summation of the potential due to each of these points (and the acceleration is ΔU)

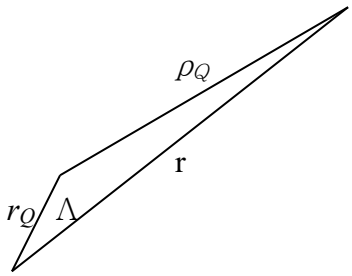
$$U = G\sum_{Q=1}^{\infty}\frac{m_Q}{\rho_Q} \quad \xleftarrow[\text{same form}]{} \quad U = \frac{\mu}{r}$$

  The change in potential due to an infinitesimal element of mass $dm_\oplus$ is

$$dU = G\frac{dm_\oplus}{\rho_Q}$$

$$U = G\int_{body}\frac{1}{\rho_Q}dm_\oplus$$

  Let $r = \sqrt{x^2 + y^2 + z^2}$ and $r_Q = \sqrt{\xi^2 + \eta^2 + \zeta^2}$



Here Λ can be determined from the inner product relation:

$$\vec{r}\cdot\vec{r}_Q = rr_Q\cos\Lambda$$

$$\Rightarrow \cos\Lambda = \frac{\vec{r}\cdot\vec{r}_Q}{rr_Q} = \frac{x\xi + y\eta + z\zeta}{rr_Q} \tag{8-10}$$

The second cosine law from the planar trigonometry leads to

$$\rho_Q^2 = r^2 + r_Q^2 - 2rr_Q\cos\Lambda = r^2\left[1 + \frac{r_Q^2}{r^2} - \frac{2rr_Q}{r^2}\cos\Lambda\right]$$

$$\Rightarrow \rho_Q = r\sqrt{1 - \frac{2r_Q}{r}\cos\Lambda + \left(\frac{r_Q}{r}\right)^2} = r\sqrt{1 - 2\alpha\cos\Lambda + \alpha^2},\ \alpha \equiv \frac{r_Q}{r}$$

from which the potential becomes

$$U = G\int_{body}\frac{1}{r\sqrt{1-2\alpha\cos\Lambda+\alpha^2}}dm_\oplus,\ \begin{cases}\cos\Lambda \le 1\\ \alpha < 1\ (P\ \text{is outside the central body})\end{cases}$$

- Series Expansion by Legendre Polynomials

  Use the binomial expansion to expand the denominator, which leads to a series of Legendre polynomials, P_l:

$$\frac{1}{\sqrt{1-2\alpha\, cos\,\Lambda+\alpha^2}} = \sum_{l=0}^{\infty}\alpha^l P_l\left[cos\,\Lambda\right] = \sum_{l=0}^{\infty}\alpha^l P_l\left[\gamma\right],\ \gamma = cos\Lambda$$

where

$$P_l[\gamma] = \frac{1}{2^l l!}\frac{d^l(\gamma^2-1)^l}{d\gamma^l}\ \text{ or }\ P_l[\gamma] = \frac{1}{2^l}\sum_{j=0}^{l}\frac{(-1)^j(2l-2j)!}{j!(l-j)!(l-2j)!}\gamma^{l-2j} \tag{8-11}$$

Examples:

$$P_0[\gamma]=1,\ P_1[\gamma]=\gamma,\ P_2[\gamma]=\frac{1}{2}(3\gamma^2-1),\ P_3[\gamma]=\frac{1}{2}(5\gamma^3-3\gamma)$$

Then, the potential can be written as

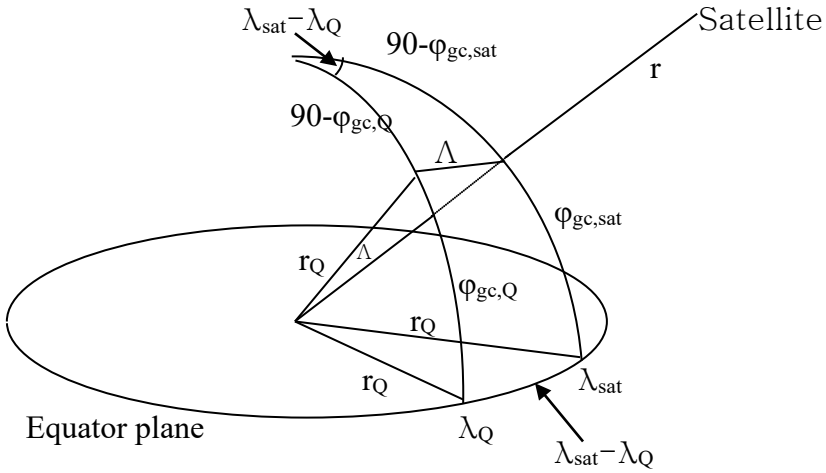
$$U = \frac{G}{r}\int_{body}\sum_{l=0}^{\infty}\alpha^{l}P_{l}\left[cos\,\Lambda\right]dm_{\oplus} \qquad (8\text{-}12)$$

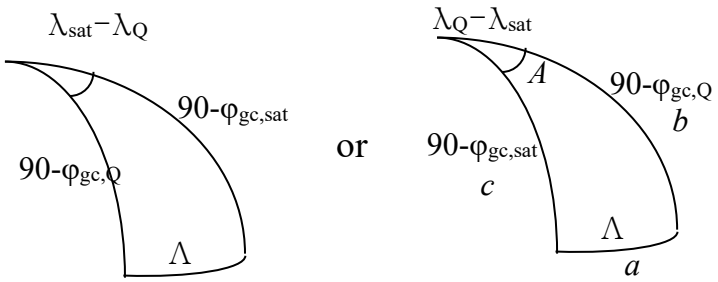
which, however, is not very useful, since we cannot directly find the angle, Λ

- Substitution of Λ as a function of known quantities by spherical-harmonic expansion

λ_Q : *longitude of Q*

ϕ_{gcQ} : *geocentric latitude of Q*





The cosine law of spherical trigonometry gives

$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

$$\cos\Lambda = \cos(90° - \phi_{gcQ})\cos(90° - \phi_{gcsat}) + \sin(90° - \phi_{gcQ})\sin(90° - \phi_{gcsat})\cos(\lambda_Q - \lambda_{sat})$$
$$= \sin\phi_{gcQ}\sin\phi_{gcsat} + \cos\phi_{gcQ}\cos\phi_{gcsat}\cos(\lambda_Q - \lambda_{sat}) \qquad (8-14)$$

The addition theorem of spherical harmonics provides a way to expand for the angle Λ (Hobson, E. W., 1965, The Theory of Spherical and Ellipsoidal Harmonics, New York, Chelsea publishing, p141)

$$P_l[\cos\Lambda] = P_l[\sin\phi_{gcQ}]P_l[\sin\phi_{gcsat}] + 2\sum_{m=1}^{l}\frac{(l-m)!}{(l+m)!}\left[A_{l,m}A'_{l,m} + B_{l,m}B'_{l,m}\right] \quad (8\text{-}15)$$

where

$$A_{l,m} = P_{l,m}[\sin\phi_{gcQ}]\cos(m\lambda_Q)$$
$$A'_{l,m} = P_{l,m}[\sin\phi_{gcsat}]\cos(m\lambda_{sat})$$
$$B_{l,m} = P_{l,m}[\sin\phi_{gcQ}]\sin(m\lambda_Q)$$
$$B'_{l,m} = P_{l,m}[\sin\phi_{gcsat}]\sin(m\lambda_{sat})$$
$$P_{l,m}[\gamma] = \frac{1}{2^l l!}(1-\gamma^2)^{m/2}\frac{d^{l+m}(\gamma^2-1)^l}{d\gamma^{l+m}} = (1-\gamma^2)^{m/2}\frac{d^m}{d\gamma^m}P_l[\gamma] \quad (8\text{-}16)$$

$P_{l,m}$ (l: degree, m: order) is the associated Legendre functions

Introducing (8-15) into (8-12) leads to

$$U = \frac{G}{r}\int_{body}\sum_{l=0}^{\infty}\left(\frac{r_Q}{r}\right)^l\left[P_l[\sin\phi_{gcQ}]P_l[\sin\phi_{gcsat}] + 2\sum_{m=1}^{l}\frac{(l-m)!}{(l+m)!}(A_{l,m}A'_{l,m} + B_{l,m}B'_{l,m})\right]dm_\oplus \quad (ii)$$

- Final Form by (1) isolating Terms which depend only on the central body and (2) non-dimensioanlization:

When m $\geq$ 1, Define

$$C'_{l,m} = \int_{body} r_Q^l\ 2\frac{(l-m)!}{(l+m)!}A_{l,m}dm_\oplus$$
$$= \int_{body} r_Q^l\ 2\frac{(l-m)!}{(l+m)!}P_{l,m}[\sin\phi_{gcQ}]\cos(m\lambda_Q)dm_\oplus$$
$$S'_{l,m} = \int_{body} r_Q^l\ 2\frac{(l-m)!}{(l+m)!}B_{l,m}dm_\oplus \quad (8\text{-}17)$$
$$= \int_{body} r_Q^l\ 2\frac{(l-m)!}{(l+m)!}P_{l,m}[\sin\phi_{gcQ}]\sin(m\lambda_Q)dm_\oplus$$

These coefficients in (8-17) represent the mathematical modeling for the Earth's shape using spherical harmonics

- The coefficient $C'_{l,0}$ (m=0) is called the zonal harmonics

$$C'_{l,0} \equiv \int_{body} r_Q^l\ P_l[\sin\phi_{gcQ}]dm_\oplus$$

- The coefficient $S'_{l,0}$=0, since $m = 0 \Rightarrow \sin\left(m\lambda_Q\right) = 0$

Then, the potential function can be rewritten as

$$U = \frac{G}{r}\sum_{l=0}^{\infty}\frac{P_l[\sin\phi_{gcsat}]}{r^l}C'_{l,0} + \frac{G}{r}\sum_{l=1}^{\infty}\sum_{m=1}^{l}\frac{P_{l,m}[\sin\phi_{gcsat}]}{r^l}\left[C'_{l,m}\cos(m\lambda_{sat}) + S'_{l,m}\sin(m\lambda_{sat})\right] \quad (iii)$$

Non-dimensionalize the gravitational coefficients by scaling with the Earth's means equatorial radius $\left(R_\oplus\right)$ and mass $\left(m_\oplus\right)$:

$$C'_{l,m} = C_{l,m}R_\oplus^l m_\oplus, \quad S'_{l,m} = S_{l,m}R_\oplus^l m_\oplus$$

Then, *(iii)* becomes

$$\begin{aligned} U &= \frac{\mu}{r}\sum_{l=0}^{\infty}P_l[\sin\phi_{gcsat}]\left(\frac{R_\oplus}{r}\right)^l C_{l,0} \\ &+ \frac{\mu}{r}\sum_{l=1}^{\infty}\sum_{m=1}^{l}P_{l,m}[\sin\phi_{gcsat}]\left(\frac{R_\oplus}{r}\right)^l\left[C_{l,m}\cos(m\lambda_{sat}) + S_{l,m}\sin(m\lambda_{sat})\right] \end{aligned} \quad (8\text{-}18)$$

where note that $S_{l,0} \equiv 0$

(8-18) can be expressed as

$$U = \frac{\mu}{r}\sum_{l=0}^{\infty}\sum_{m=0}^{l}P_{l,m}[\sin\phi_{gcsat}]\left(\frac{R_\oplus}{r}\right)^l\left[C_{l,m}\cos(m\lambda_{sat}) + S_{l,m}\sin(m\lambda_{sat})\right]$$

$l = 0 \sim \infty,\ m = 0 \quad \Rightarrow$ The first term of (8-18)

$l = 1 \sim \infty,\ m = 1 \sim l \quad \Rightarrow$ The second term of (8-18)

When l=0, m=0

$$\frac{\mu}{r}P_{0,0}[\sin\phi_{gcsat}]\left(\frac{R_\oplus}{r}\right)^0\left[C_{0,0} + 0\right] = \frac{\mu}{r}$$

$$\left(\begin{array}{l} \Uparrow P_{0,0}[\sin\phi_{gcsat}] = 1 \\ \Uparrow C_{0,0} = 1 \leftarrow C'_{0,0} = C_{0,0}R_\oplus^0 m_\oplus \leftarrow C'_{0,0} = \int_{body} r_Q^0\ P_{0,0}[\sin\phi_{gcQ}]dm_\oplus = m_\oplus \end{array}\right)$$

If the origin of the coordinates system coincides with the center of mass of attracting body, the coefficients $C_{1,0}, C_{1,1}$ and $S_{1,0}$, $S_{1,1}$ are all zero (Proofs are provided below)

$$U = \frac{\mu}{r}\left[1 + \sum_{l=2}^{\infty}\sum_{m=0}^{l} P_{l,m}[\sin\phi_{gcsat}]\left(\frac{R_\oplus}{r}\right)^l \left[C_{l,m}\cos(m\lambda_{sat}) + S_{l,m}\sin(m\lambda_{sat})\right]\right] \quad (8\text{-}19)$$

Let $-C_{l,0} \equiv J_l$ (8-20)

$$U = \frac{\mu}{r}\left[1 - \sum_{l=2}^{\infty} J_l\left(\frac{R_\oplus}{r}\right)^l P_l[\sin\phi_{gcsat}] + \sum_{l=2}^{\infty}\sum_{m=1}^{l}\left(\frac{R_\oplus}{r}\right)^l P_{l,m}[\sin\phi_{gcsat}]\left[C_{l,m}\cos(m\lambda_{sat}) + S_{l,m}\sin(m\lambda_{sat})\right]\right] \quad (8\text{-}21)$$

The analysis of satellite motion allows us to empirically determine the coefficients $C_{l,m}$ and $S_{l,m}$ from observations

Often use normalized coefficients; the conventional gravitational coefficients become very small as the degree and order (l,m) grow large.

$$\bar{S}_{l,m} = \Pi_{l,m} S_{l,m},\ \bar{C}_{l,m} = \Pi_{l,m} C_{l,m},\ \Pi_{l,m} = \sqrt{\frac{(l+m)!}{(l-m)!k(2l+1)}},\ \begin{cases} k = 1\ if\ \ m = 0 \\ k = 2\ if\ \ m \neq 0 \end{cases}$$

- Proof that $C_{1,0} = C_{1,1} = S_{1,1} = 0$ if the origin of the coordinate system coincides with the center of mass of attracting body

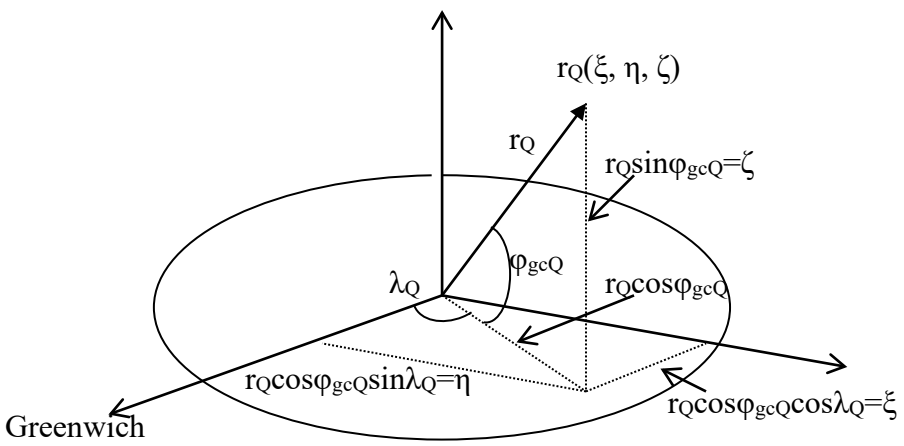
  The proof begins by recalling the definition of $\left(C'_{l,m}, S'_{l,m}\right)$

$$C'_{l,m} = C_{l,m} R_\oplus^l m_\oplus = \int_{body} r_Q^l\ 2\frac{(l-m)!}{(l+m)!} P_{l,m}[\sin\phi_{gcQ}]\cos(m\lambda_Q) dm_\oplus$$

$$S'_{l,m} = S_{l,m} R_\oplus^l m_\oplus = \int_{body} r_Q^l\ 2\frac{(l-m)!}{(l+m)!} P_{l,m}[\sin\phi_{gcQ}]\sin(m\lambda_Q) dm_\oplus$$

Note that $C'_{1,0} = C'_{1,1} = S'_{1,1} = 0 \Leftrightarrow C_{1,0} = C_{1,1} = S_{1,1} = 0$ since

$$C'_{l,m} = C_{l,m} R_\oplus^l m_\oplus, \qquad S'_{l,m} = S_{l,m} R_\oplus^l m_\oplus$$



- Proof that $C_{1,0} = 0$

$$C'_{1,0} = \int_{body} r_Q^1 \; 2\frac{1!}{1!} P_{1,0}[\sin\phi_{gcQ}] dm_\oplus$$

$$\left(\Uparrow \; P_{1,0}[\sin\phi_{gcQ}] = \sin\phi_{gcQ} \;\Leftarrow\; P_{1,0}[x] = x\right)$$

$$= 2\int_{body} r_Q \sin\phi_{gcQ} dm_\oplus$$

$$= 2\int_{body} \zeta dm_\oplus = 0 \;\leftarrow \text{if origin is the center of mass of body}$$

- Proof that $C_{1,1} = 0$

$$C'_{1,1} = \int_{body} r_Q^1 \; 2\frac{0!}{2!} P_{1,1}[\sin\phi_{gcQ}] \cos\lambda_Q dm_\oplus$$

$$\left(\Uparrow \; P_{1,1}[\sin\phi_{gcQ}] = \cos\phi_{gcQ} \Leftarrow P_{1,1}[x] = (1-x^2)^{1/2}\right)$$

$$= \int_{body} r_Q \cos\phi_{gcQ} \cos\lambda_Q dm_\oplus$$

$$= \int_{body} \xi dm_\oplus = 0 \;\leftarrow \text{if origin is the center of mass of body}$$

- Proof that $S_{1,1} = 0$

$$S'_{1,1} = \int_{body} r_Q^1 \; 2\frac{0!}{2!} P_{1,1}[\sin\phi_{gcQ}] \sin\lambda_Q dm_\oplus$$

$$\left(\Uparrow \; P_{1,1}[\sin\phi_{gcQ}] = \cos\phi_{gcQ}\right)$$

$$= \int_{body} r_Q \cos\phi_{gcQ} \sin\lambda_Q dm_\oplus$$

$$= \int_{body} \eta dm_\oplus = 0 \;\leftarrow \text{if origin is the center of mess of body}$$

8.6.1.2 Spherical Harmonics

- Recall the Aspherical Gravitational Potential (8-21)

$$U = \frac{\mu}{r}\left[1 - \sum_{l=2}^{\infty} J_l \left(\frac{R_\oplus}{r}\right)^l P_l[\sin\phi_{gcsat}] + \sum_{l=2}^{\infty}\sum_{m=1}^{l}\left(\frac{R_\oplus}{r}\right)^l P_{l,m}[\sin\phi_{gcsat}]\left[C_{l,m}\cos(m\lambda_{sat}) + S_{l,m}\sin(m\lambda_{sat})\right]\right]$$

- Zonal Harmonics
  - Defined by zero-th order (*m=0*)
  - Potential field is symmetrical about the polar axis
  - Can give the mass distribution with respect to the Latitude, i.e., they are simply bands of latitude
    - For any P_l, there are *l* circles of latitude along which P_l is zero
    - Hence (*l*+1) zones in which the function P_l is alternatively increasing (+) and decreasing (-)
  - Cannot give the mass distribution with respect to Longitude, i.e., dependence of the potential on Longitude vanishes

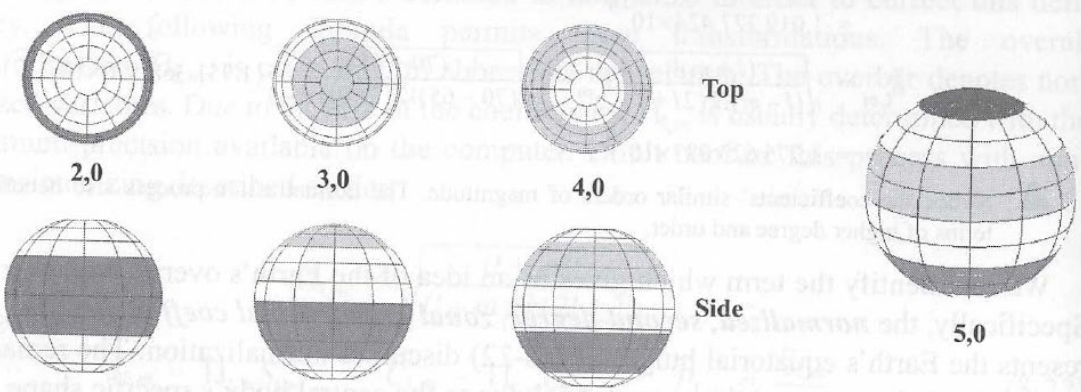


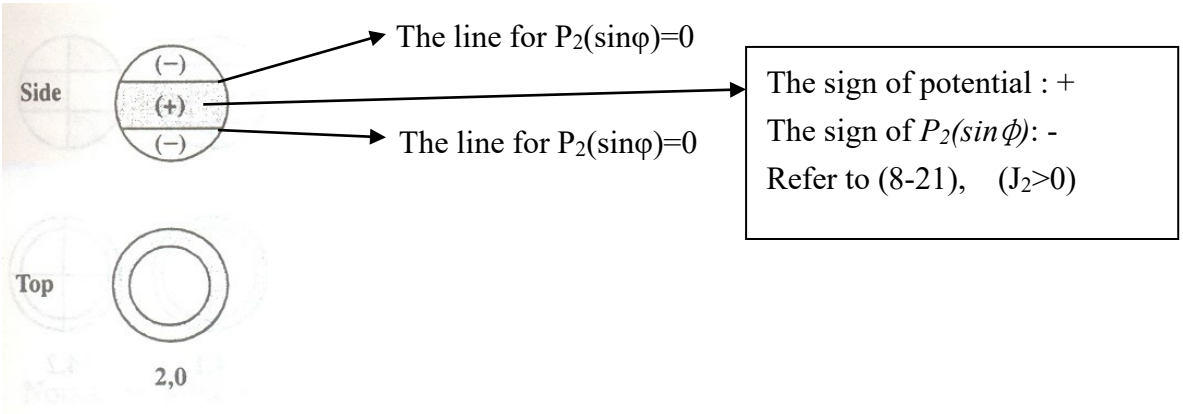
Figure 8-4. **Zonal Harmonics.** J_2 accounts for most of the Earth's gravitational departure from a perfect sphere. This band (and others) reflects the Earth's oblateness. The shading indicates regions of additional mass and the numbers link regions between the views. The third harmonic appears similar to the J_2 from the top but is reversed for the bottom view. J_5 is shown in a perspective view.

■ l=2, $P_{2,0} \equiv P_2$

$$P_2(x) = \frac{1}{2}(3x^2 - 1) = 0 \qquad \Rightarrow \textit{two solutions } x_1 \textit{ and } x_2$$

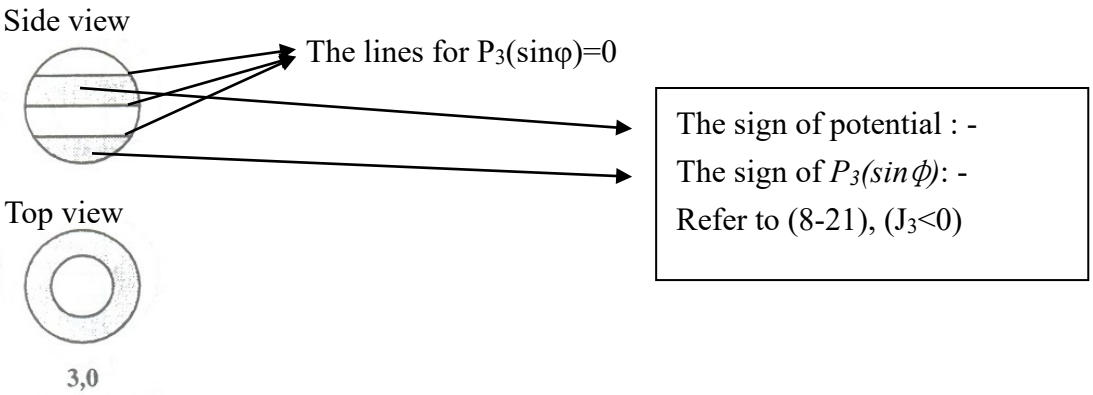
$$P_2(\sin\phi) = \frac{1}{2}(3\sin^2\phi - 1) = 0 \Rightarrow \textit{two solutions } \phi_1 \textit{ and } \phi_2$$

For the shape of the Earth, J_2 is 1000 times larger than the next largest coefficient J_3



■ l=3, $P_{3,0} \equiv P_3$

$$\mathrm{P}_3(\mathrm{x}) = \frac{1}{2}(5\mathrm{x}^3 - 3\mathrm{x}) = 0,\ \mathrm{x} = \sin\phi \ \Rightarrow\ \text{three solutions } \phi_1 = 0,\ \phi_2, \phi_3$$



- Sectorial Harmonics
  - Occur when $l=m$ and represent bands of longitude

$$P_{1,1}(x) = (1-x^2)^{1/2},\ P_{2,2} = 3(1-x^2),\ P_{3,3} = 15(1-x^2)^3 \ldots.$$
$$P_{1,1}(\sin\phi) = (1-\sin^2\phi)^{1/2},\ P_{2,2} = 3(1-\sin^2\phi),\ P_{3,3} = 15(1-\sin^2\phi)^3 \ldots.$$

$$P_{1,1} = \cos(\phi_{gc,sat}),\ P_{2,2} = 3\cos^2(\phi_{gc,sat}),\ P_{3,3} = 15\cos^3(\phi_{gc,sat}) \ldots.$$

  - Polynomials $P_{l,l}$ are zero only at the poles ($\phi = \pm 90°$), since $P_{l,l}$ is a function of cosine
  - $\sin(l\lambda)$ and $\cos(l\lambda)$ are zero for $2l$ different values of λ
  - The lines along which the functions $\sin(l\lambda)$ and $\cos(l\lambda)$ times $P_{l,l}$ vanish are meridians of longitude, which divide the sphere into $2l$ "orange slice" sectors
  - Each sector represents l positive (+) and l negative (-) mass concentrations

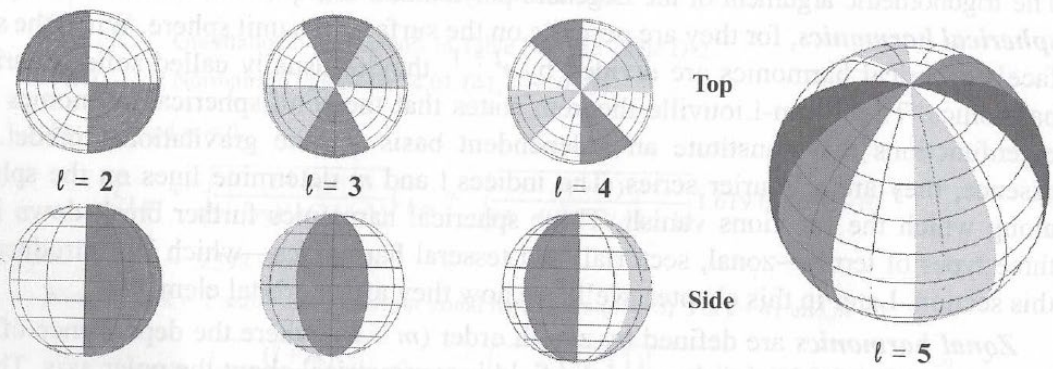
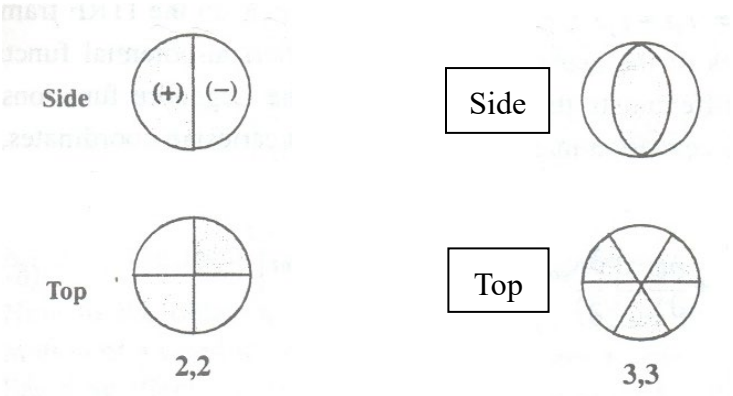


Figure 8-5. **Sectorial Harmonics.** Sectorial harmonics take into account the extra mass distribution in longitudinal regions.

  - $l=m=2,\ \ x = \sin\phi$

$$P_{2,2}(x) = 3(1-x^2) = 0 \ \Rightarrow\ x = \pm 1 \ \Rightarrow \textit{two solutions}\ \phi = \pm 90°$$

$$C_{2,2}\cos(2\lambda) + S_{2,2}\sin(2\lambda) = 0$$
$$\Rightarrow \frac{\sin(2\lambda)}{\cos(2\lambda)} = -\frac{C_{2,2}}{S_{2,2}} \Rightarrow \tan(2\lambda) = -\frac{C_{2,2}}{S_{2,2}} \Rightarrow \textit{four solutions}\ \lambda_1 \cdots \lambda_4$$



- $l=m=3,\ \ x=\sin\phi$

$$P_{3,3}(x)=15(1-x^2)^{3/2}=0\ \Rightarrow\ x=\pm 1\ \Rightarrow\ \phi=\pm 90^\circ$$

$$C_{3,3}\,cos(3\lambda)+S_{3,3}\,sin(3\lambda)=0$$

$$\Rightarrow \frac{sin(3\lambda)}{cos(3\lambda)}=-\frac{C_{3,3}}{S_{3,3}}\Rightarrow tan(3\lambda)=-\frac{C_{3,3}}{S_{3,3}}\Rightarrow six\,solutions\ \lambda_1\cdots\lambda_6$$

- Tesseral Harmonics
  - Occur when $l\neq m\neq 0$ and represent specific regions on the Earth
  - Sphere is divided up into a checkerboard array
  - The number of circles of latitude along which $P_{l,m}$ is zero is (l-m)
  - The terms $\sin(m\lambda)$ and $\cos(m\lambda)$ vanish along 2m meridians of longitude

- *l=3, m=1*

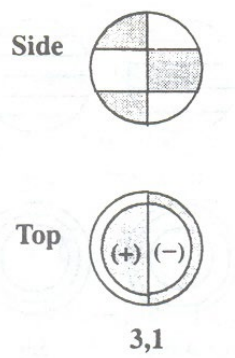
$$P_{3,1}(x) = \frac{3}{2}(5x^2 - 1)(1 - x^2)^{1/2} = 0$$

$$\Rightarrow (1 - x^2)^{1/2} = 0 \rightarrow x = \pm 1 \;\; \rightarrow \;\; \phi = \pm 90°$$

$$\Rightarrow (5x^2 - 1) = 0 \;\; \rightarrow 2(= l - m)\, solutions\ \phi_1\ and\ \phi_2 \;\; for\ P_{3,1}(x) = 0$$

$$C_{3,1}\cos(\lambda_{sat}) + S_{3,1}\sin(\lambda_{sat}) = 0$$

$$\Rightarrow \frac{\sin(\lambda_{sat})}{\cos(\lambda_{sat})} = -\frac{C_{3,1}}{S_{3,1}} \Rightarrow \tan(\lambda_{sat}) = -\frac{C_{3,1}}{S_{3,1}} \Rightarrow 2m = 2\, solutions\ \lambda_1\ and\ \lambda_2$$



- *l=2, m=1 ??*

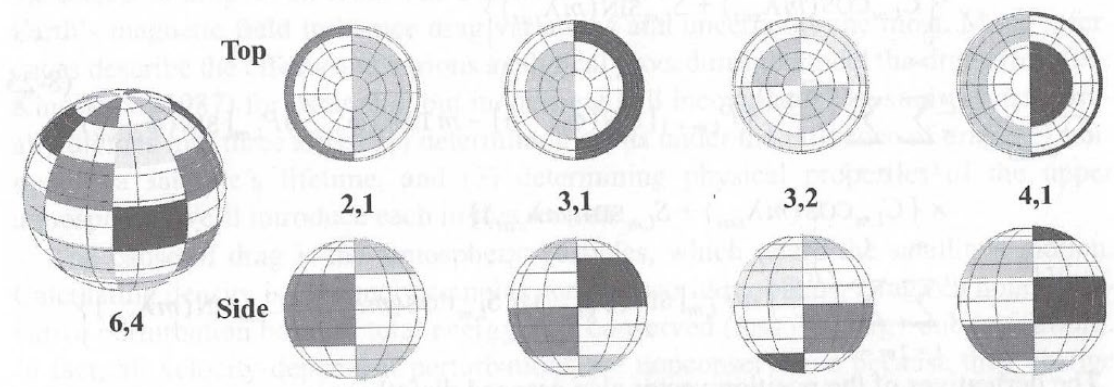


Figure 8-6. **Tesseral Harmonics.** Tesseral harmonics attempt to model specific regions on the Earth which depart from a perfect sphere. Depicting these and tying to the gravitational coefficients can be difficult.